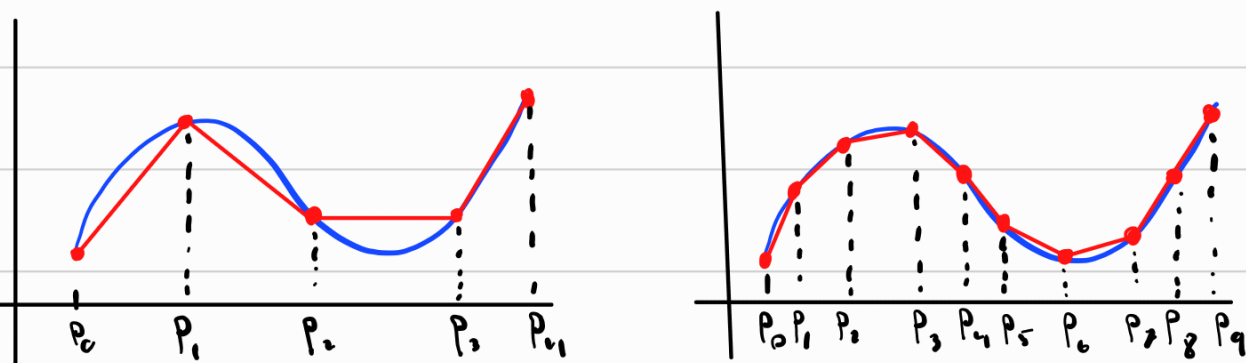


When we talk about arc length we want to know exactly how "long" a curve is. That is, if a curve is a bent wire, how long would the wire be if straight.

Let C be a curve defined by $y=f(x)$, with f continuous.

We can make a crude approximation to the arc length by drawing secant lines across the curve. More secant lines gives a better approximation



To make things easier, let's fix $\Delta x = |P_i - P_{i+1}|$ to be the same for all P_i . That is, each point is evenly spaced on the x-axis.

We then see the y-coordinate is changing. Let's look

at the distance formula for P_i to P_{i+1}

$$D = \sqrt{(P_{i+1} - P_i)^2 + \underbrace{(f(P_{i+1}) - f(P_i))^2}_{\Delta y_i}} = \sqrt{\Delta x^2 + \Delta y_i^2}$$

$f(P_{i+1}) - f(P_i)$ is denoted Δy_i as the change in y -coordinate

We then simplify

$$D = \sqrt{\Delta x^2 + \Delta y_i^2} = \sqrt{\Delta x^2 \left(1 + \frac{\Delta y_i^2}{\Delta x^2}\right)} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \cdot \Delta x$$

So arc length is

$$L \approx \sum \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \Delta x$$

If we take $\Delta x \rightarrow 0$ for infinitely small intervals, this will perfectly approximate arc length (think about how we did Riemann sums then integration).

In this case, $\Delta x \rightarrow dx$ and $\frac{\Delta y}{\Delta x} \rightarrow \frac{dy}{dx}$ $\nearrow$ a function of x

So, if $\frac{dy}{dx}$ is continuous on $[a, b]$, the arc length of a curve $y = f(x)$, $a \leq x \leq b$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Note: If $x = f(y)$, we similarly have

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Ex: Find Arc Length of $y^2 = x^3$ between the points $(1, 1)$ and $(4, 8)$.

This is quadrant 1, here the curve is fully described by

$$y = x^{3/2}$$

then

$$\frac{dy}{dx} = \frac{3}{2} x^{1/2}$$

with $1 \leq x \leq 4$

So, arc length is

$$L = \int_1^4 \sqrt{1 + \left(\frac{3}{2}x^{1/2}\right)^2} dx$$

$$= \int_1^4 \sqrt{1 + \frac{9}{4}x} dx, \quad \text{let } u = 1 + \frac{9}{4}x$$
$$du = \frac{9}{4}$$

$$= \frac{4}{9} \int_{\frac{13}{4}}^{10} \sqrt{u} du$$

$$= \frac{4}{9} \left. \frac{2}{3} x^{3/2} \right|_{\frac{13}{4}}^{10}$$

$$= \frac{8}{27} \left(10^{3/2} - \frac{13^{3/2}}{4} \right)$$

$$= \frac{8}{27} \left(\sqrt{1000} - \sqrt{\frac{2197}{64}} \right)$$

$$= \frac{8}{27} \left(10\sqrt{10} - \frac{13}{8}\sqrt{13} \right)$$

$$= \frac{1}{27} (80\sqrt{10} - 13\sqrt{13})$$

$$L \approx 7.634$$

We can think of arc length as a function of x

Observe

$$S(x) = \int_a^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

is arc length of some function from a to x .

Notice

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

These formulas become useful for surface area.